

Creating Event Forwarding on Splunk App for SOAR Export app on Splunk Enterprise (Search Head)

Summary

This document explains in detail the steps to create and configure an Event Forwarder in the Splunk App for SOAR Export app of Search Head to map the fields received from alert SPL into Standard CEF Fields before Sending to SOAR. Also documents steps to Ensure the connection of SOAR with Splunk Enterprise and Validate the mapped fields are reaching SOAR. This document also features an explanation about the Design considerations and Field Mapping approach for better understanding of the steps.

1. Open Splunk App for SOAR Export app

In Search Head UI,

Go to `Apps > Splunk App for SOAR Export`

2. Create an Event Forwarding

- 1) Under **Event Forwarding** tab,
Click **Add New**

2) Do the following:

i) Select **Saved Search Export****Name:** PrivEsc_Forwarding**Saved Search:** Select the newly created alert from the drop-down (Ex: Privilege Escalation - Domain + Local Admin Detection)**Artifact Label - Optional:** privilege_escalation_event**Select Destination:** Select the SOAR instance that you are connected to from the drop-down (Ex: SOAR)**Container Name - Optional:** Select Add a non-standard field > member_sid**Container Label - Optional:** privilege_escalation**Schedule****Select Every**

5 minutes

Earliest Time: -10m**Latest Time:** now > Next

Add NewStep 1 of 2

Data Model Export

Saved Search Export

Name
PrivEsc_Forwarding

Saved Search
Privilege Escalation - Domain + Local Admin Detecti...

Artifact Label - Optional
privilege_escalation_event

Select Destination
SOAR

Container Name - Optional
member_sid

Container Label - Optional
privilege_escalation

Schedule

☐ Real Time

☒ Every

5

 Minutes

Earliest Time

-10m

Latest Time

now

Cancel

Next

ii) **Severity and Sensitivity Fields****Severity:** High**Sensitivity:** TLP:Amber

Add NewStep 2 of 2

Configuring [Privilege Escalation - Domain + Local Admin Detection](#) on **SOAR**

Severity and Sensitivity Fields

Severity ?

High

Sensitivity ?

TLP: Amber

iii) Mapped Fields

Note: You can either create a Custom (non-CEF) field under Search Fields and add a non-standard field under CEF Fields. Here I have used:

Search Fields > Custom (non-CEF) field

CEF Fields > used the existing standard CEF fields so that it can be mapped easily on the SOAR playbook.

When you create this initially, you will map every Search Field under **Unmapped Fields Section**. Later when the Event Forwarding is triggered, the Search Fields are automatically populated under Mapped Fields Section.

Search Fields

actor
EventCode
group_name
member_sid
source
system
user

CEF Fields

sourceUserName
externalId
cs1
destinationUserId
cs2
deviceHostname
destinationUserName

Click Save and Close.

Add New
Step 2 of 2

▼
Unmapped Fields (0)

Group ?	Search Fields	CEF Fields	Contains	
☐	actor	sourceUserName	user name	-
☐	EventCode	externalId	Select contains	-
☐	group_name	cs1	Select contains	-
☐	member_sid	destinationUserId	Select contains	-
☐	source	cs2	Select contains	-
☐	system	Choose the CEF field n...	Select contains	-
☐	user	destinationUserName	user name	- +

☐ Save mappings ?

>
Mapped Fields (0)

Cancel
Previous
Save and Preview
Save and Close

3. Design Considerations and Field Mapping Approach

During the design of this use case, event data was mapped directly to Common Event Format (CEF) fields within the **Splunk App for SOAR Export** rather than creating additional custom artifacts within Splunk SOAR.

Reason for Using Event Forwarding Field Mapping

The Splunk App for SOAR Export provides **native support** for forwarding search results into **standard CEF fields**. This approach offers several advantages:

- Simplifies integration between Splunk and SOAR.
- Reduces playbook complexity.
- Ensures consistent field availability across artifacts.
- Allows filtering, decision logic, and enrichment actions to use standardized field names.
- Minimizes additional parsing requirements within SOAR.

Mapped Fields

Splunk Field	CEF Field
member_sid	destinationUserId
added_user	destinationUserName
actor	sourceUserName
group_name	cs1
source	cs2
system	deviceHostname
EventCode	externalId

Alternative Approach Considered

An alternative design would have been to create custom artifact fields within SOAR and populate them during playbook execution.

This approach was not selected because:

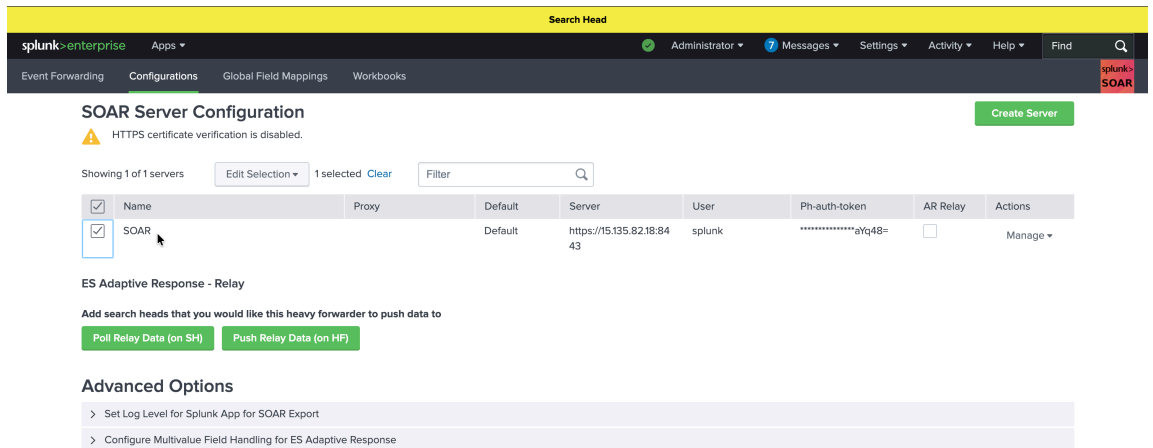
- Additional playbook logic would be required.
- Increased maintenance complexity.
- More difficult troubleshooting.
- Reduced portability of the playbook.

Using standardized CEF field mappings provided a simpler and more scalable implementation for this project.

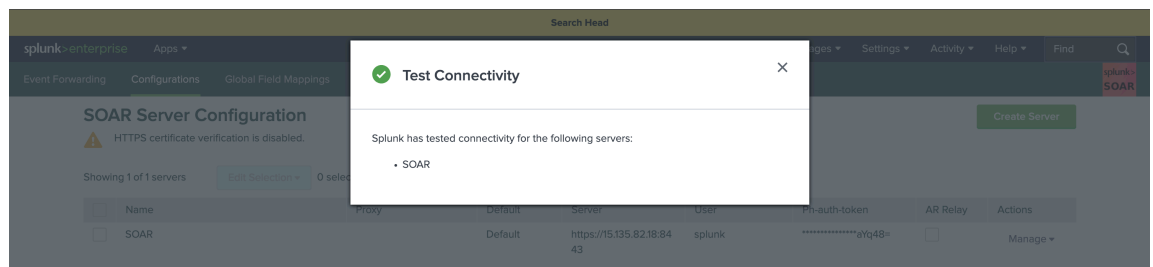
4. Test Event Forwarding execution

1) Test Connectivity with SOAR

- i) In **Splunk App for SOAR Export** app on **Search Head UI**,
Go to **Configurations** tab,
Look for the **SOAR** instance > **Manage** > **Edit**
Ensure that the **Server:** `https://<SOAR_PublicIP>:8443` > **Save**



- ii) Click on **Manage** on the **SOAR** instance in the table > **Test Connectivity**
Should see a **green tick** visible.



2) Preview the Event Forwarding and Send to SOAR

Go to **Event Forwarding** tab,

Click on newly created Forwarder > Next > Save and Preview

The events will be visible if the alert is triggered properly.

Click on **Send to SOAR**

Preview for PrivEsc_Forwarding

Preview window

Last hour

Your search matched the following data

Successfully parsed CEF fields

Container Name

S-1-5-21-3021654831-3058487264-1625118876-1125

Sensitivity

amber

Severity

high

cs1

Administrators

cs2

Client

destinationUserId

S-1-5-21-3021654831-3058487264-1625118876-1125

deviceHostname

EC2AMAZ-CLIENT01.splunkadmin.local

externalId

4732

sourceUserName

admin

Send to SOAR

Your search matched the following data

Successfully parsed CEF fields

Container Name

S-1-5-21-3021654831-3058487264-1625118876-1125

Sensitivity

amber

Severity

high

cs1

Domain Admins

cs2

DC

destinationUserId

S-1-5-21-3021654831-3058487264-1625118876-1125

destinationUserName

eric wan

deviceHostname

EC2AMAZ-DC01.splunkadmin.local

externalId

4728

sourceUserName

admin

Send to SOAR

Done

5. Verify on Splunk SOAR

Login to **SOAR UI as admin**,

The event should be visible on the **Dashboard**.

Click on that and Events page will open

Click on one event and expand.

Expected: You should see all the artifacts that was created on Event Forwarding here with data.

The screenshot displays the Splunk SOAR interface. At the top, the 'splunk SOAR' header is visible. Below it, the 'ARTIFACTS (1)' section shows a table with one artifact. The artifact details are expanded, revealing a list of mapped fields and their values. The bottom section shows 'NOTES (1)' with a search bar and a list of notes.

ID	LABEL	NAME	SEVERITY	CREATED BY	TAGS
529	privilege_escalation...	S-1-5-21-3021654831-3058487264-1625118876-1125	High		

Details:

- Name: S-1-5-21-3021654831-3058487264-1625118876-1125
- Label: privilege_escalation_event
- Description: (PrivEsc_Forwarding) added by Splunk App for SOAR Export
- Source ID: 83c5ab269315dfac0cd53f0a0c5b108
- Severity: High

Mapped Fields:

- _originating_search: http://SearchHead8000/app/search/search?q=%7C+savedsearch+%22Privilege+Escalation+++Domain%2B+Local+Admin+Detection%22&latest=1778818468
- sa1: domain Admin
- sa2: DC
- destinationUserId: S-1-5-21-3021654831-3058487264-1625118876-1125
- destinationUserName: DC
- deviceHostname: EDDAMAZ-DC01 splunkadmin.local
- externalId: 4728
- sourceUserName: admin

NOTES (1):

- Privilege Escalation Detected

This **confirms** Event Forwarding is connected to SOAR, and the mapped fields are available for CEF Field mapping inside the SOAR Playbook.

Next Steps

Create a SOAR Playbook to automate a response for the Privilege Escalation attack – Refer [27_Creating_Privilege_Escalation_SOAR_Playbook_V1.pdf](#).